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Development of a pig wean-quality score using machine-learning algorithms to characterize and classify groups with high mortality risk under field conditions

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ABSTRACT

Mortality during the post-weaning phase is a critical indicator of swine production system performance, influenced by a complex interaction of multiple factors of the epidemiological triad. This study leveraged retrospective data from 1723 groups of pigs marketed within a US swine production system to develop a Wean-Quality Score (WQS) using machine learning techniques. The study evaluated three machine learning models, Random Forest, Support Vector Machine, and Gradient Boosting Machine, to classify groups having high or low 60-day mortality, where high mortality groups represented 25 % of the groups among the study population with the highest mortality values (n=431; 60-day mortality=9.98 %), and the remaining 75 % of the groups were of low mortality (n=1292; 60-day mortality=2.75 %). The best-performing model, Random Forest (RF), outperformed the other ML models in terms of accuracy (0.90), sensitivity (0.84), and specificity (0.92) metrics, and was then selected for further analysis, which consisted of creating the WQS and ranking the most important factors for classifying groups as high or low mortality. The most important factors ranked through the RF model to classify groups with high mortality were pre-weaning mortality, weaning age, average parity of litters in sow farms, and PRRS status. Additionally, stocking conditions such as stocking density and time to fill the barn were important predictors of high mortality. The WQS was developed and correlated ($r = 0.74$) with the actual 60-day mortality of the groups, offering a valuable tool for assessing post-weaning survivability in swine production systems before weaning. This study highlights the potential of machine learning and comprehensive data utilization to improve the assessment and management of weaned pig quality in commercial swine production, which producers can utilize to identify and intervene in groups, according to the WQS.

1. Introduction

The swine industry is pivotal in the ever-growing global demand for high-quality protein (Mote and Rothschild, 2020). Swine mortality during the post-weaning phase is a crucial indicator of sustainability for commercial production systems, where increased survivability can significantly improve production efficiency and profitability. However, achieving optimal performance and sustainability in swine production systems requires a comprehensive understanding of the factors influencing post-weaning mortality. A review on post-weaning mortality indicated that almost 10 % of the weaned pigs produced in the U.S. swine industry do not reach the market due to post-weaning mortalities (Gebhardt et al., 2020a).

The complex dynamics of post-weaning mortality involve an intricate interplay of host, pathogen, and environmental factors, requiring a holistic approach to characterize and manage post-weaning mortality effectively. Also, the relationship among these variables is dynamic, shifting over time and between different production environments. Therefore, understanding and quantifying these influences is essential for swine producers to make informed decisions and allocate resources

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efficiently.

For optimal performance and the realization of the genetic potential in commercial pigs during their growth phase, the quality of weaned pigs, which is impacted by health and performance factors (e.g., age, weight, and disease status), is pivotal (Bruns and Stalder, 2019; Fix et al., 2010). Multiple studies conducted under field conditions and utilizing data from commercial farms reported the importance of the pre-weaning phase of production, which included risk factors related to the conditions in which weaned pigs are stocked to growing sites, on the downstream post-weaning mortality (Gebhardt et al., 2020a, 2020b). Altogether, these factors are related to the quality of the weaned groups, which is crucial for achieving the genetic potential of commercial pigs starting in the growing phase, along with optimal conditions related to the epidemiological triad until the animals reach market age.

The evaluation of the quality at weaning of weaned pigs is often done subjectively, without a specific quality score or index, that considers characteristics related to the animals when weaned, such as weaning age, weight, health statuses, and the conditions when placed in growing sites. These metrics are often utilized to understand the background of the pigs without transforming them into measurable metrics that may support the decision-making process.

In recent years, advances in data science and machine learning (ML) have offered a powerful toolkit for analyzing and harnessing the vast amounts of data generated within the livestock realm (Berckmans, 2017; García et al., 2020), and specifically in the swine industry (Akinyemi et al., 2023; Benjamin and Yik, 2019). The utilization of machine-learning algorithms to create a score index that is predictive of events related to animal production, for example, was reported in other studies that used ML to create biosecurity scores (Silva et al., 2019) or movements and spatial data (Machado et al., 2019; Silva et al., 2019) to predict disease outbreaks. However, applying ML models on large datasets to develop a metric related to a wean-quality score of pigs still needs to be developed.

Therefore, the objective of this study was to utilize retrospective data from a large number of groups of pigs marketed within a large U.S. swine production system to train multiple machine learning models and create a wean-quality score for groups of pigs that was utilized to classify groups based on their likelihood of having high or low mortality rates during the post-weaning phase. For this purpose, diverse information from the characteristics of the groups` was collected and utilized by the ML models, where the predictive accuracy of the Wean-Quality-Score was evaluated by employing three distinct machine-learning models, and the best model was selected to create the Wean-Quality-Score and rank the most important factors for accurately classifying groups as high or low mortality for the initial 60-days in the wean-to-finish using the variables included in the model.

2. Materials and methods

2.1. Overview and study design

This retrospective observational study utilized data from 1754 closeouts of pigs marketed between January 2022 and April 2023 from one US swine production system in the midwestern region.

The sample population for this study was defined as the individual groups of pigs (n=1754) marketed within a midwestern US swine producer, the source population was all the closeout groups marketed in this company, and the target population was swine production systems with similar conditions of production. The outcome variable of the analysis was the average 60-day mortality of each closeout, representing the initial 60 days in the wean-to-finish (W2F) phase, and the explanatory variables were characteristics concerning the groups` retrospective performance in the pre-weaning phase and at placement in growing sites (aka. stocking conditions). The outcome 60-day mortality was converted into a dichotomous variable (High vs. Low 60-day mortality) by categorizing the top 25 % (n=441) of groups with the highest 60-day mortality averages as "High", while the remaining groups (75 %; n=1313) were classified as "Low", even though the "Low" category groups included the remaining portion of the data that included the low and median mortality groups. The cutoff point for this study population to separate the two categories was 5.35 %.

SAS® Version 9.4 (SAS Institute, Inc., Cary, NC) algorithms were built to combine different data streams containing information on the closeouts in the wean-to-finish phase, creating a closeout-specific longitudinal data set, here referred to as master-table. The historical data on the master-table was utilized in this study for training and testing the performance of three different machine learning (ML) models on classifying the groups as having High or Low mortality based on the retrospective performance and health in the pre-weaning phase along with conditions at placement in growing sites. The best model was then selected for ranking the factors and creating a wean-quality metric based on the predictive value from the ML model.

2.2. Characteristics of the data streams

Historical data for each group of pigs, here defined as the observational unit of analyses, was imported automatically from the producer using Power Automate (Microsoft Corporation, Redmon, WA, USA) to create an automated data workflow between the company database and our local storage server, where the data streams were processed using SAS algorithms developed to conduct the data wrangling, merging, and cleaning processes. The data utilized in this study represented five different data streams: sow farm (aka. breeding herd) health status and productivity data, finisher closeout reports, growing pig health status, and infrastructural and management reports. The data imported for this study was then integrated into a single master-table, for analysis using SAS software, utilizing a data management procedure described before (Magalhães et al., 2022). After completing data aggregation, data validation was performed by removing any closeouts without complete data for all twelve predictor variables and negative or missing values for the outcome 60-day mortality. Lastly, variable selection was conducted using the Boruta features reduction algorithm (Kursa and Rudnicki, 2010) to measure if the initial number of predictor variables (n=12) used in this study to classify groups` mortality could be reduced.

2.3. Characteristics of the predictor variables

From the data streams mentioned above and integrated into a master-table, twelve variables hypothesized to be important to influence the quality of weaned pigs were selected as predictors to be utilized in the establishment of the wean-quality score, to assess productivity and health conditions of the weaned piglets, and condition at their placement in growing sites. Table 1 describes the characteristics of each variable used to estimate the wean quality score. Completeness of the data was assessed, and 30 groups were removed due to missing values, resulting in 1723 groups.

2.4. Machine learning algorithms

Once the predictor variables were selected from the final master-table, three machine learning (ML) algorithms were applied to the data set to investigate the predictive capability of each ML model on accurately classifying the groups as high or low 60-day mortality based on the twelve predictors concerning the historical performance from breeding-to-stocking. The ML models utilized in this study were Random Forest (RF) (Breiman, 2001), Support Vector Machine (SVM) (Boser et al., 1992), and Gradient Boosting Machine (GBM) (Friedman and Meulman, 2003).

Before comparing the performance of the ML models, a variable reduction was conducted using the Boruta features reduction algorithm (Kursa and Rudnicki, 2010), which selects relevant features by measuring the importance of each variable using a tree-like algorithm,

Table 1
Description of the predictor variables included in the final RF model.

Variable Category*	Low mortality (n=1292)	High Mortality (n=431)	Overall (n=1723)
Wean pig medication[a]			
Enrofloxacin	577 (44.7 %)	114 (26.5 %)	691 (40.1 %)
Tulathromycin	305 (23.6 %)	215 (49.9 %)	520 (30.2 %)
Ceftiofur	30 (2.3 %)	2 (0.5 %)	32 (1.9 %)
Mixed	128 (9.9 %)	62 (14.4 %)	190 (11.0 %)
No medication	201 (15.6 %)	27 (6.3 %)	228 (13.2 %)
Florfenicol	51 (3.9 %)	11 (2.6 %)	62 (3.6 %)
Sow farm medication[b]			
Chlortetracycline	412 (31.7 %)	105 (24.4 %)	515 (29.9 %)
Lincomycin	139 (10.8 %)	69 (16.0 %)	208 (12.1 %)
Mixed	59 (4.6 %)	31 (7.2 %)	90 (5.2 %)
No medication	684 (52.9 %)	226 (52.4 %)	910 (52.8 %)
Ventilation type[d]			
Natural	1145 (88.6 %)	378 (87.7 %)	1523 (88.4 %)
Tunnel	147 (11.4 %)	53 (12.3 %)	200 (11.6 %)
Number of sources			
One	833 (64.5 %)	294 (68.2 %)	1127 (65.4 %)
Two	402 (31.1 %)	113 (26.2 %)	515 (29.9 %)
Three	57 (4.4 %)	24 (5.6 %)	81 (4.7 %)
***M. hyopneumoniae* status**			
Endemic	650 (50.3 %)	291 (67.5 %)	941 (54.6 %)
Mixed	139 (10.8 %)	49 (11.4 %)	188 (10.9 %)
Negative	505 (38.9 %)	91 (21.1 %)	594 (34.5 %)
PRRS virus status			
Endemic	799 (61.8 %)	332 (77.0 %)	1131 (65.6 %)
Epidemic	13 (1.0 %)	57 (13.2 %)	70 (4.1 %)
Mixed Status	1 (0.1 %)	0 (0 %)	1 (0.1 %)
Negative	479 (37.1 %)	42 (9.7 %)	521 (30.2 %)
Pre-weaning mortality Mean in % (SD)[c]	12.9 % (2.7)	14.1 % (5.4)	13.2 % (3.6)
Parity at farrow Mean parity (SD)[c]	4.07 (0.8)	4.01 (0.7)	4.06 (0.8)
Weaning age Mean days (SD)[c]	18.5 (2.1)	18.7 (2.6)	18.6 (2.2)
Time to fill Mean days (SD)[c]	18.3 (6.3)	19.7 (6.4)	18.6 (6.3)
Stocking density[e] Mean proportion (SD)[c]	1.93 (0.5)	2.09 (0.3)	1.97 (0.4)

* Variables' categories
[a] Medication in piglets for treatment purposes;
[b] Medication in sows for treatment purposes;
[c] Continuous Variable value followed by the standard deviation (SD);
[d] Both categories have ventilation systems using fans, where the "natural" type is also known as curtain barn type
[e] Calculated by dividing the total pig capacity of the site at placement (i.e., starting-age pigs) by the total number of pigs stocked.

where an all-relevant method is utilized rather than a minimal optimal approach. After this step, eleven variables remained on the final model.

Initially, the complete dataset was randomly divided into two quasi-independent portions, here referred as training and test sets. The training set included 80 % of the original data, and the test set accounted for the remaining 20 %. Ten repetitions of a 10-fold cross-validation procedure on the training set was conducted for each ML model, where performance was estimated by predicting the mortality as high or low for the observations included in one of the folds of the training set, while using the remaining folds for the training the model. This process is then repeated ten times until the whole training set is utilized in the cross-validation procedure.

The cross-validation procedures were conducted in R for all ML models. The hyperparameter tuning for each model was specifically configured to identify optimal parameters as follows: the Random Forest (RF) model tuned the 'mtry' parameter (number of predictors) across a grid of values (3−8); the Support Vector Machine (SVM) testing of different combinations of parameters (sigma and C for SVM with radial kernel) was performed over a grid determined by the caret package; and the Gradient Boosting Machine (GBM) tuned the depth and number of trees, and minimum observations in terminal nodes.

After the cross-validation procedure, the performance of each ML model was assessed using the test set (20 % of the original dataset) and *bestTune* was utilized to select the tuning parameters from the cross-validation. For this purpose, the accuracy of the ML models in classifying the groups' mortality was measured using a confusion matrix (Table 2), which is a classical table constructed to define the performance of classification models (Machado et al., 2015) and is commonly utilized to measure the correct number of positive or negative, in this case, high or low, outcomes (Chadaga et al., 2021). For each ML model, the results reported in the confusion matrix represented an average of the performance of each ML model on the test data, after the 10-fold cross-validation procedure.

The performance of each ML model reported in the confusion matrix was assessed by calculating accuracy, sensitivity, and specificity, similar to other studies (Machado et al., 2019, 2015; Silva et al., 2019) as recommended by Dohoo et al. (2003). Accuracy represented the proportion of groups correctly predicted with high (positive) or low (negative) 60-day mortality, while specificity was calculated by dividing the true negative groups by the sum of true negative and false positives, and sensitivity by dividing the true positive groups by the sum of true positives and false negatives. The results for each ML model represent the average performance across all folds.

The data was modeled for the three ML algorithms using two approaches: the first utilized the complete dataset with the natural imbalance between high and low 60-day mortality groups, and a second approach using a down-sampling method, where the number of low mortality groups was matched down to the number of high mortality groups. In fact, no difference in performance was demonstrated by the two approaches. Thus, we utilized the complete dataset in this study with its naturally imbalanced distribution.

2.5. Selection and ranking of variables by relative importance

After comparing the performance of the three ML models in terms of accuracy, sensitivity, and specificity, the model with the best results for these parameters was selected and the following steps applied: (1) variable ranking, and (2) partial dependence plots.

Once the best model was selected, a confusion matrix is drawn and variable ranking (1) was conducted by calculating the Gini index represented on the variable importance plot, which characterizes the relevance of each wean quality factor included in the model in accurately classifying groups with high or low 60-day mortality, where higher values represent more important predictors. Friedman and Meulman, (2003) introduced the concept of "relative importance," which is employed to assess the significance of predictor variables. This estimation is based on the number of times a variable is selected through the RF model for splitting, weighted by the squared improvement of the models resulting from each split, and then averaged across all trees.

After identifying and ranking the variables using the best ML model, partial dependence plots (2) for each predictor were plotted to understand the patterns within each variable related to the prediction of high or low mortality. The partial dependence plot measures the variables' relevance and provides information related to the marginal estimates within each factor utilized in the final model, which provides a graphical representation of the level of dependence on the prediction. Given that we designed our analysis as a binary classification for case or control (i.e., high or low 60-day mortality), we reported partial dependence values on the 'logit' scale (Cutler et al., 2007), where greater values signify an increased probability of belonging to a case herd.

Table 2
Overall performance of the ML models on classifying the groups' 60-day mortality in the test data.

Performance Parameter*	Machine Learning Model		
	RF	*SVM*	*GBM*
ACC (95 % C.I.)	0.9070 (0.8646, 0.9306)	0.8140 (0.7719, 0.8563)	0.8663 (0.8257, 0.9004)
Se	0.8462	0.6964	0.7500
Sp	0.9248	0.8368	0.9015
PPV	0.7674	0.4535	0.6977
NPV	0.9535	0.9341	0.9225

* Performance on the ML models on the complete and unbalanced dataset after removing PRRS vaccine variable; ACC (95 % C.I.): accuracy and 95 % confidence interval; Se=Sensitivity; Sp=Specificity; PPV=Positive predictive value; NPV= Negative predictive value. RF=Random Forest; SVM=Support Vector Machine; GBM= Gradient Boost Machine.

2.6. Development and validation of a wean-piglet quality score

The positive predictive value (PPV) and negative predictive value (NPV) for each observation was computed, where PPV and NPV refer to the probability of being classified as high or low 60-day mortality, respectively, which was calculated using the probability from the trained RF model for each group being classified as high (positive) or low (negative). The probabilities were extracted from the prediction of the test set using the RF model. In summary, PPV represents the probability (from zero to one) that a group predicted as high 60-day mortality had a 60-day mortality above the 5.35 % cutoff, while NPV is the probability of accurately classifying low mortality groups as low mortality (McKenna and Dohoo, 2006). For creating the score, we focused on the NPV obtained for each group, which was multiplied by ten, thus converting the original NPV of each group into a Wean-Quality-Score (WQS), which ranged between zero and ten, thus facilitating the interpretation of the results. After creating the WQS for each group (n = 1723), a linear regression model was fitted to assess the correlation between the actual 60-day mortality and the NPV of each group in terms of coefficient of correlation (R). In other words, this step was performed to validate further and estimate the meaning of the WQS.

3. Results

3.1. Descriptive analysis

The final master-table contained breeding-to-market information on data from 1764 W2F groups (~3,900,000 pigs) from six data streams and 41 sow farms. The mean 60-day mortality for all groups of pigs (n=1764) was 4.56 % (95 % confidence interval (CI) 4.36 %-4.76 %). The mean 60-day mortality for the 25 % of the groups (n=431) with the highest mortality among the study population was 9.98 %, while the remaining 75 % (n=1292) had 2.75 % average mortality.

3.2. Model selection and validation

When comparing the performance of the three ML models in terms of accuracy, sensitivity, and specificity, RF outperformed the other models (SVM and GBM), as described in Table 2, and, therefore, used for further analyses. The Boruta feature reduction algorithm was performed, where one variable (PRRSV vaccine at weaning) was identified to be removed from the analysis, resulting in eleven predictors included in the final analysis. The performance of the three ML models was tested again, where we observed an overall increase in terms of accuracy for all models after removing the variable identified by the Boruta procedure (PRRS vaccine at weaning).

3.3. Variable selection and ranking according to relative importance

After selecting RF as the best ML model for classifying high and low mortality groups (Table 2), the contribution of the eleven variables included in the model for accurately classifying the groups as high or low 60-day mortality was assessed by ranking the variables according to their relative importance using the Gini index (Fig. 1). Across the eleven factors ranked using the RF model, the most important factors were weaning age, pre-weaning mortality, stocking density, average parity at farrow, time to fill the growing site, antibiotic application at weaning, and PRRS status at weaning. The least four important were sow farm medication type, *M. hyopneumoniae* status, number of sow farm origins, and ventilation type of growing site.

Partial dependence plots were generated to demonstrate the effect of each of the eleven variables used on the RF ML model on accurately classifying groups with high or low 60-day mortality, highlighting the importance of the levels within each variable by demonstrating the marginal effect patterns.

Fig. 2 displays the plots for the most important variables identified in the variable importance plot. The y-axis results for each plot represents the probability of predicting groups as positive (in this case, high 60-day mortality) given a specific value across the x-axis after controlling for the remaining covariates present in the model. For example, lower negative values for the partial dependence plot of lactation days are observed when the value is closer to 10 days, indicating that high 60-day mortality values are more likely to be predicted in this x-axis range.

Overall, the following patterns on the partial dependence plots, for the seven most important variables indicated in the variable importance plot, were associated with groups classified as having high mortality by the RF model: young weaning age in weaned groups, high pre-weaning mortality, high stocking density, young and old (extreme values) average parity at farrow, growing sites with long fill days, weaned pigs that were treated with antibiotics based on Tulathromycin, and groups originating from sow farms with epidemic status for PRRS.

3.4. Establishing the wean-quality score

When creating the wean-quality score, the negative predicted values (NPV) were estimated for each group of pigs, and these values were correlated with the actual 60-day mortality of the groups on their original format (continuous variables) before classifying them as high or low. The Pearson correlation coefficient estimated between the actual mortality and NPV for all groups was 0.74 (Pearson r = 0.74).

Fig. 3 describes the relationship between the NPV values for each group and the 60-day mortality of the groups, where it can be observed visually that groups with a wean-quality score below two are clustered with higher mortality values, while groups with a score above eight are clustered on the opposite side (i.e., the higher the wean-quality score of the groups, the lower the downstream mortality for the groups.

4. Discussion

Swine mortality occurring during the post-weaning phase is an important indicator of performance concerning the sustainability of a swine production system. This event is caused by multiple factors involving the epidemiological triad of host, pathogen, and environment, and often, this interaction changes over time within one environment

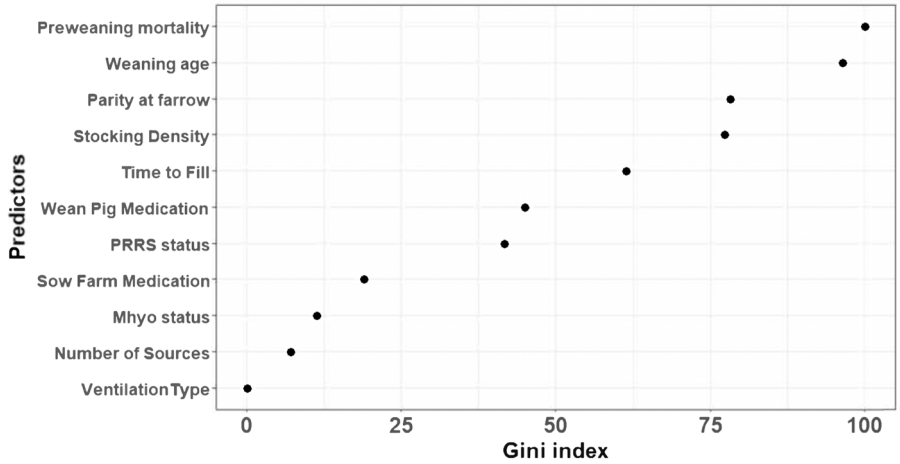

Fig. 1. : Variable relative importance analysis using the RF model for the set of predictor variables.

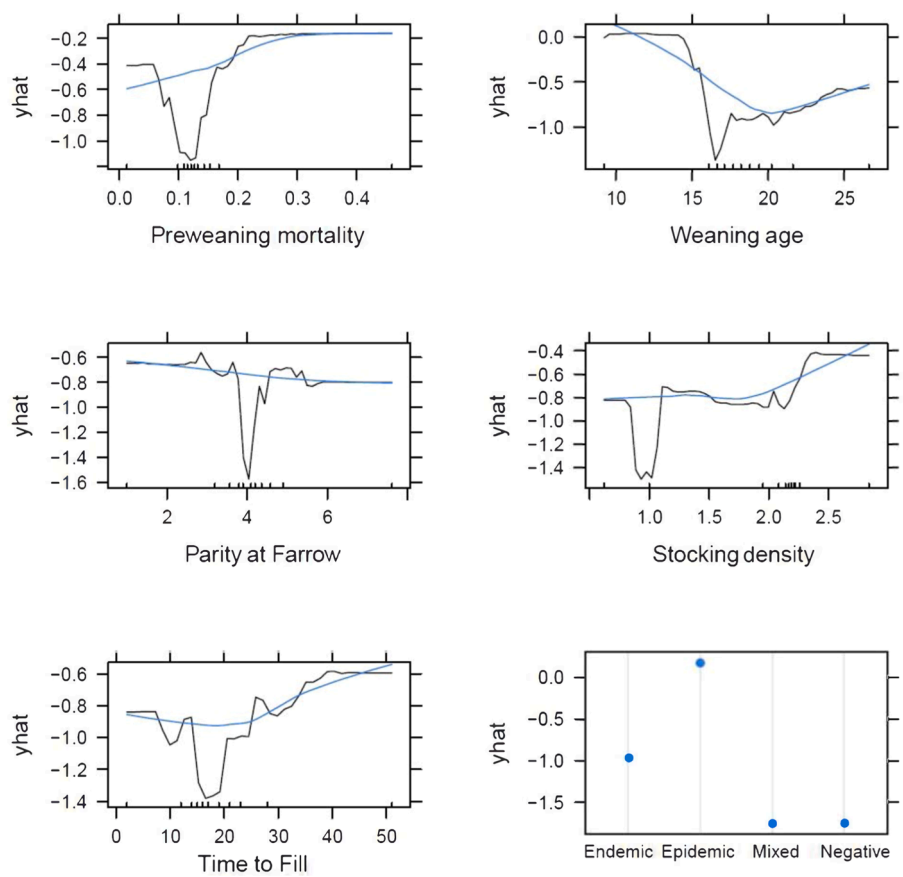

Fig. 2. : Partial dependence plots containing the marginal effect for high 60-day mortality. The y-axis results for each plot represent the probability of predicting groups as high 60-day mortality given a specific value across the x-axis after controlling for the remaining covariates present in the model. The y-axis values are computed in relation to the High mortality category, where values closer to zero indicate higher probability of being classified as High mortality.

and between farms.

The quality of weaned pigs is a key component for achieving the maximum genetic potential of commercial pigs starting in the growing phase, along with optimal conditions related to the epidemiological triad until the animals reach market age. Multiple observational studies conducted under field conditions and utilizing data from commercial farms reported the importance of the pre-weaning phase of production and factors related to the conditions at placement in growing sites (Gebhardt et al., 2020b).

This study demonstrated the importance of utilizing retrospective information from commercial groups of pigs within a large swine production system in the U.S. Midwestern region to characterize mortality during the initial 60 days of the post-weaning phase-related weaning factors. The most important variables identified by the RF machine learning model in the study combined sow farm-related factors that are often utilized to characterize batches of weaned pigs flowing from sow farms, such as average pre-weaning mortality, weaning age, average parity of the litters that farrowed the pigs, and status for PRRS, which is

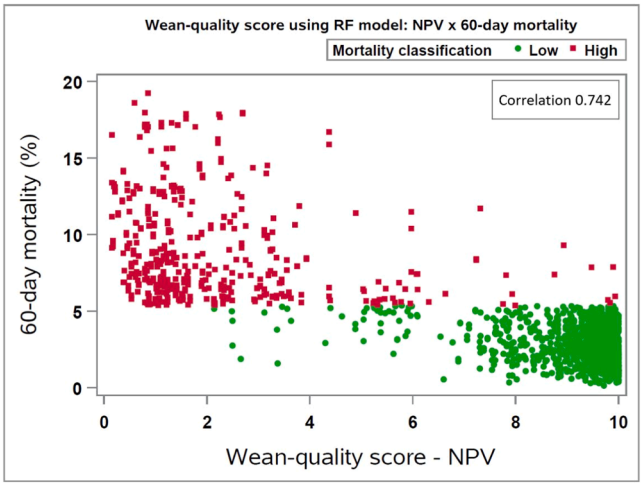

Fig. 3. : Correlation plot between the wean-quality score and the 60-day mortality, as well as the classification as high or low mortality categories. The majority of the low 60-day mortality closeouts are clustered in the area of the chart with the best wean-quality scores.

one of the major pathogens impacting modern swine industry.

Pre-weaning mortality was the most important factor across all variables. However, this variable indicates the overall performance of the groups of weaned pigs throughout the pre-weaning phase, which lasts ~ 21 days. Previous authors demonstrated the application of the pre-weaning mortality percentage as a proxy of performance for the pigs (e.g., higher mortality associated with decreased low birth weight (Declerck et al., 2016; Pardo et al., 2013; Škorput et al., 2018; Smith et al., 2007; Zotti et al., 2017), poor management conditions (Calderón Díaz et al., 2017), and animals born from gilt litters (Miller et al., 2012),

The results of the present study also demonstrated an increased probability of higher 60-day mortality for groups with lower weaning ages, which was consistent with other studies (Davis et al., 2006; Leliveld et al., 2013; López-Vergé et al., 2019; Losinger et al., 1998; Main et al., 2004). When optimal weaning age and appropriate weight are achieved, the pig's functional maturity is achieved. Furthermore, the parity of the sows-producing piglets has been demonstrated to be associated with increased post-weaning mortality (Craig et al., 2017; Piñeiro et al., 2019), especially for offspring litters from gilts litters that have been inferior milk and colostrum quality (Declerck et al., 2016). This pattern was also observed through the partial dependence plots from this study, which indicated a higher probability for high 60-day mortality when the groups of pigs originated from young parity sow farms.

Across all potential pathogens circulating in sow farms that may impact swine health, PRRS virus has been reported to impact greatly the downstream post-weaning performance of weaned pigs in sow farms undergoing an outbreak (Alvarez et al., 2015; Dong et al., 2018; Holtkamp et al., 2013; Lyoo et al., 2015; Magalhães et al., 2022; Tian et al., 2007; Trevisan et al., 2020). This pattern was also observed in the present study, where the RF model ranked sow farm PRRS status at weaning for the groups of pigs as an important factor. Also, groups classified as PRRS epidemic on the partial dependence plot (i.e., the epidemic phase of infection in a breeding herd) had a higher probability of being predicted as high 60-day mortality groups compared to the remaining categories.

Lastly, variables characterizing the stocking conditions of weaned pigs in the growing site were important predictors for high mortality in this study, where groups of pigs placed with high stocking density, from multiple breeding herd sources, and with longer time to fill the barn had a higher probability for being classified as high 60-day mortality groups. These patterns were also identified in previous studies, where longer fill days create larger variation within a group for the age and weight of the pigs (Dewey et al., 2006), and multiple sources are associated with larger mortality rates (Agostini et al., 2015, 2014; Maes et al., 2004). This study considered pre-weaning and stocking conditions factors to create the WQS as groups of pigs may have low performance due to either factor alone or both simultaneously (e.g., good quality of weaned pigs in terms of pre-weaning phase factors, but placed under poor environmental conditions).

Even though several studies are available in the scientific literature concerning risk factors of post-weaning survivability related to the pre-weaning phase and stocking conditions, to the best of the authors' knowledge, this is the first study that consolidated field data from commercial groups of growing pigs to create a wean-quality score and measure its predictive performance for classifying groups with high or low mortality. Conversely, multiple articles are available in non-peer-reviewed literature referring to the importance of the quality of weaned pigs, but the industry lacks an actual measurable score. This study demonstrated the importance of comprehensive utilization of swine big data, along with standalone artificial intelligence technologies such as the ML algorithm hereby utilized to create a wean-quality score, or index, that can be used for many approaches, such as for classifying mortality of pigs used this study, but not limited to other outcomes (e.g., average daily weight gain, feed conversion, or full value pig). Also, the wean-quality score utilized in this study could be utilized for ongoing and automated prediction of mortality in groups of weaned pigs, as most of the variables utilized in the present study are collected as part of the routine of swine farms. The high correlation between the Wean-Quality Score and the actual 60-day mortality demonstrated to be a practical tool to evaluate the quality of weaned pig groups at placement in growing sites and its capability to predict high mortality groups (i.e., 60 days prior).

The machine learning applied in the present study for forecasting mortality are commonly used in other studies in the livestock realm demonstrating their application for predicting important KPIs of productivity (Aiken et al., 2020; Alonso et al., 2013; Golden et al., 2019; Kamphuis et al., 2010; Machado et al., 2015; Murphy et al., 2014; Nguyen et al., 2020; Shine et al., 2018; Silva et al., 2019; Zhang et al., 2019).

The major limitation of this study is related to its external validity. Despite the good internal validity in this study for the system providing the data, its extrapolation to other companies should be avoided if not accounting for expected differences between swine companies, swine populations, locations, and time frames. For instance, the conditions of the population used in this study reflect pigs raised in high dense areas in the US, and, consequently, under higher health challenge scenario.

Furthermore, the quality of the information is prone to information bias as the information is recorded on-site and then exported to data management software, a process that may be prone to typographical errors. Thus, the magnitude of the effect for each variable should not be utilized in the decision-making process. For example, if the objective is to measure the impact of weaning age on downstream mortality, and specific causal diagram should be developed and causal inference algorithms such as propensity score implemented (Rosenbaum and Rubin, 1983).

This study is also limited to the data recorded and then provided from the production system, not including, for example, key performance indicators, such as birth weight (Declerck et al., 2016; Pardo et al., 2013; Škorput et al., 2018; Smith et al., 2007; Zotti et al., 2017), cross-fostering (Calderón Díaz et al., 2017), colostrum intake, daily room temperature, and other management factors. Collecting these parameters could provide more precise information related to the batches of pigs weaned from sow farms. However, despite being key characteristics to be collected, recording this type of information is not a common procedure as part of the routine in breeding herds due to labor or, when collected, would represent a small portion of the overall population on the farm. Also, data on management procedures and housing conditions that are different between the groups throughout the growing phase were not accounted in this study, thus, not investigated as potential

predictors of mortality. Therefore, there is a great opportunity to improve the characterization of the WQS if more representative information starts to be collected, along with the current metrics, as part of the routine of swine production systems.

Altogether, the potential of combining comprehensive data management along with machine-learning was demonstrated in this study, where the WQS and the ranking of factors predictive of high mortality groups could support the decision-making process and improve the management and decision-making process for weaned pigs in commercial swine production.

5. Conclusions

The machine-learning pipeline described in this study provides an automated and structured method to create a Wean-Quality Score (WQS) for commercial groups of pigs, to classify groups of pigs as having either high or low 60-day mortality rates. The Random Forest (RF) model was chosen for developing the WQS, which provided an accuracy of 0.90 for accurately classifying groups with high 60-day mortality, and obtained high correlation ($r = 0.74$) with the mortality of the groups. The most influential factors in predicting high 60-day mortality included pre-weaning mortality, weaning age, average parity of litters in sow farms, and PRRS status, stocking density, and time to fill the barn.

The development of the Wean-Quality Score represents a novel approach to assessing post-weaning survivability in swine production systems. The score, correlated with actual 60-day mortality, offers a practical tool for producers to evaluate the quality of weaned pig groups. The study demonstrates the potential of machine learning and the comprehensive utilization of data to enhance the management of weaned pig quality in commercial swine production. Also, this study demonstrates the importance of applying algorithms to data collected under field conditions to characterize groups of pigs from large datasets, according to the wean-quality of groups placed in growing sites. Notably, this analysis helps producers prioritize the most important factors contributing to swine post-weaning mortality which may be used to guide the allocation of resources to reduce mortality.

CRediT authorship contribution statement

Gustavo S. Silva: Writing – review & editing, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Daniel Linhares:** Writing – review & editing, Visualization, Validation, Resources, Project administration, Funding acquisition. **Chong Wang:** Writing – review & editing, Validation, Supervision, Data curation. **Will Lopez:** Writing – review & editing, Visualization, Investigation, Conceptualization. **Derald Holtkamp:** Writing – review & editing, Visualization, Methodology, Investigation, Conceptualization. **Giovani Trevisan:** Writing – review & editing, Methodology, Investigation, Conceptualization. **Cesar Moura:** Writing – review & editing, Visualization, Resources, Investigation, Conceptualization. **Danyang Zhang:** Writing – review & editing, Validation, Resources, Investigation, Data curation. **Edison Magalhaes:** Visualization, Validation, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of Competing Interest

The authors declare no conflicts of interest.

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